



Lab notes: Intelligent cage

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Github repo: <https://github.com/DCC-Lab/IntelligentCage>

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1 Abstract for Navigate

Drug testing and disease research, among other types of experiments, require animal models to improve our fundamental biological understanding of the disease and verify the drug viability. Daily mice weighting is sometimes required to assess the animal's health. However, measuring the animal's weight and evaluating the animal's behaviour, while essential, require the interaction of an experimenter, which could change the animal's behaviour by inducing stress and jeopardizing its wellbeing. Also, the whole process is time-consuming for the experimenter, preventing continuous data acquisition. Evaluating the behaviour in the animal's home cage, as well as continuously following the food intake, the water intake and the animal's weight, could provide more accurate information on its state of mind and habits. Automated mouse home cage systems already exist on the market, but they are not customizable and they are too expensive to be supplied in multiple copies for biological experiments done with many animals. In this study, we present an automated mouse home cage system that is easily adaptable to any home cage, low-cost and user-friendly for non-experts in electronics and programming. Our goal is to reduce the human interaction and obtain more data on the animal's behaviour. In addition, our approach complements any other already existing optical monitoring systems, such as video tracking, infrared beams and radio-frequency identification. We take advantage of an Arduino for its ease of implementation, cost-effectiveness and versatility. To weigh the animals, a load cell is attached to two 3D-printed platforms of the size of the experimenter's choice, which costs less than 20\$CAD in total to make. When an animal is on the platform, its weight is measured according to the deformation of the load cell, which is evaluated using a Wheatstone bridge of four strain gauges/deformation-sensitive resistors. With a calibration function, the Arduino converts the digital signal into a value of weight, which is then stored for real-time assessment and future analysis. A tiling of multiple load cells with their platform is placed in the mouse home cage to weigh the mouse at any time and at any location. The tiling is also used to assess the mouse movements in its home cage, which can indicate signs of hyperactivity or stress behaviour. The size of the platforms are variable; smaller platforms necessitate more load cells to cover the whole cage area, but also provides better resolution in the mouse movement. Our system also measures the real-time mouse location with a tiling of capacitive sensing plates. The aluminium plate acts as a half-capacitor which can be completed by the animal's body. Its presence on the plate influences the capacity, which is linked to the charging time of the capacitor that can be easily measured. Also, we are currently working on estimating the water consumption with a loadcell to measure the variation of weight of the water bottle through time. This method will be coupled with a capacitive touch sensor at the tip of the water bottle to count the number of drinking events. We present a simple automated mouse home cage system that can be easily reproduced in any laboratory with non-expensive electronics and our open-source script.

2 Drift of TAL 221 Load cell 100g

2.1 Characterization of the drift while empty for a few hours, June 5 2023

Goal Determine the behaviour of the drift of the TAL 221 Loadcell 100g during the heating up of the circuit to determine a way to correct it.

Motivation The Intelligent Cage is meant to be used for many consecutive days. The weight measurements need to be precise and constant throughout the entire experiment because we want to track the evolution of the weight. A non-corrected drift would biased the measurements and the conclusions. That's why it's important to characterize it before going further in conception.

Methods A TAL 221 Loadcell 100g was used. The one used was bought on amazon (link at the end of the section). It was mounted on a 3D printed support (see picture). The loadcell was connected to a HX711 loadcell amp (link at the end) to filter and amplify the analog signal before reading it with an Arduino Uno.

The Arduino library HX711 by Bodge was then used to perform a measurement using the code on the github repo: <https://github.com/DCC-Lab/IntelligentCage>. To look at the drift, every reading of the output of the HX711 was stored in an micro SD connected the Arduino Uno. The MicroSD Module V1.0 DFRobot.com was used to write the data on the output on the micro SD card on a csv file.

The acquisition lasted for about 2 hours before the Arduino stopped for an unknown reason. 9000 readings were taken during that delay, so a sampling frequency of approximately 1 reading/s. IMPORTANT: The data acquisition started at the same time that we turned on the Arduino and the circuit. The effect of the heating up process was what we wanted to see.

Python was then used to apply a low-pass filter on the data to reduce noise that was important. Only the first 1/25 of the frequencies were kept. Since the drift is a slow variation, no important information was cut.

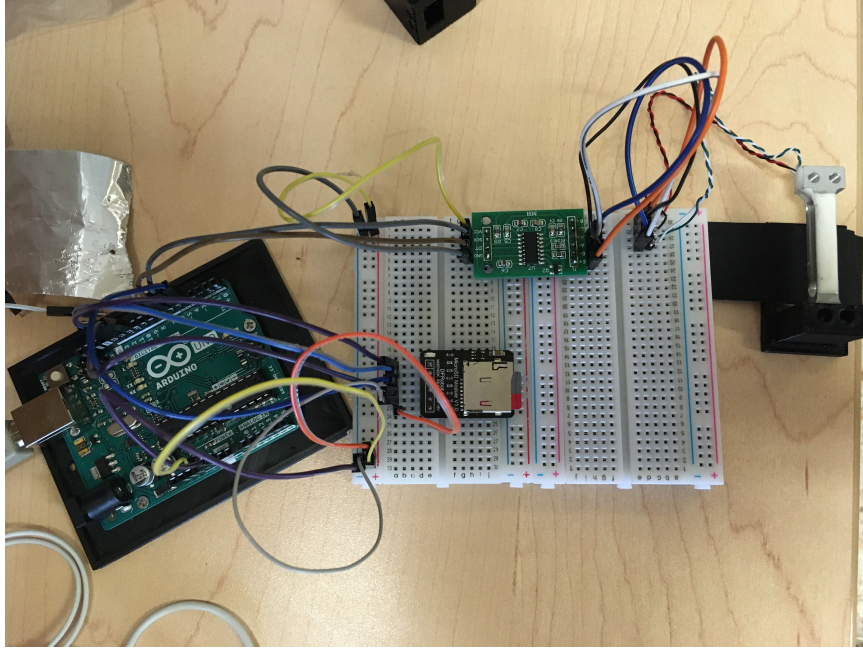


Figure 1: circuit built of Arduino Uno(left), HX711 amp (center top), micro SD module (center bottom), TAL 221 loadcell (left)

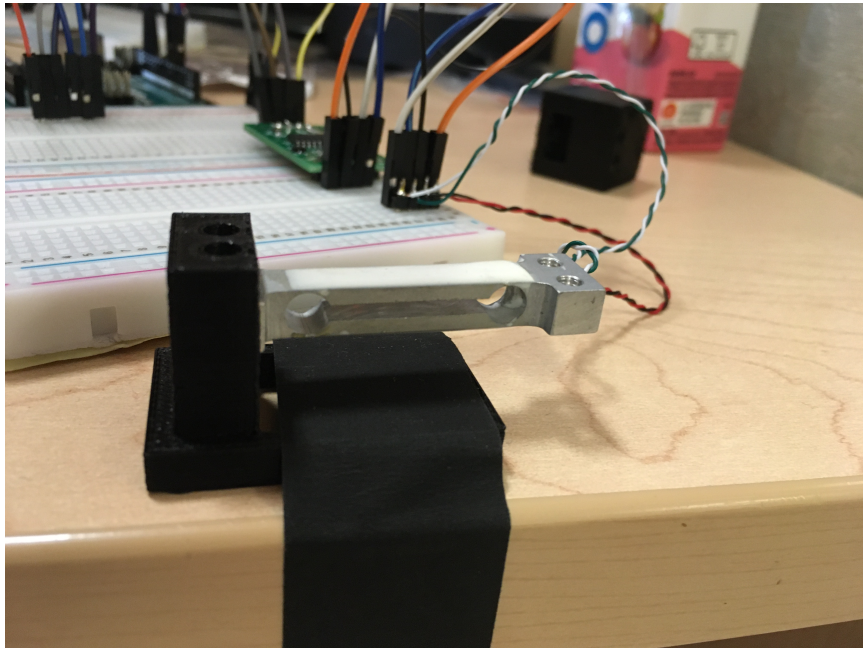


Figure 2: The TAL221 loadcell and the 3D printed support

Results To look at the drift, the output of the loadcell was first graphed to see the behaviour. The signal was very noisy (probably because only 1 reading was averaged for each loop), so a low-pass filter was used to cut the noisy high frequencies. These figures were obtained.

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Amazon link of HX711: https://www.amazon.ca/-/fr/Stemedu-modules-pr%C3%A9cision-amplificateur-Raspberry/dp/B07MTYT95R/ref=sr_1_5?keywords=HX711&qid=1684251116&sr=8-5

2.2 Characterization of the drift of weight measurements over night with a load, June 14, 15 and 16 2023

Goal Characterize the drift of the weight measurements for a longer period than in previous experiment while the arduino is powered with an external power supply, not a computer.

Motivation Same as experiment 1, but in a closer to real applications context. A bigger base and platform for the loadcell were used, closer to what will be used during real experiments. Also, the first experiments was done on raw output of the loadcell which was very noisy. We looked at weight values for this experiment. This is a more representative characterization of the drift because looking at weight values requires to calibrate the loadcell, the determination of a tare offset and a scale factor (slope).

In the previous experiment where raw output for an empty loadcell was observed, only the drift of the tare offset was observed. Since we need to calculate weight from raw output, a scale coefficient needs to be determined at the start of the experiments. The drift will be caused by the tare offset like in the previous experiment but also by the scale coefficient.

Methods The same electrical components from the first experiments were used. A different base and platform were used. They were bigger and more stable which is more likely to be similar to what will be used. The same electronic circuit was used [1](#). The difference between this experiment and the first one is that we wanted to read mass values from raw output. To do this, calibration of the tare offset and the scale coefficient was necessary. A reference weight of 37 g was used for calibration, an Arduino Mega (why not). The following table shows the tare offset and the scale factor for both night of June 14, 15 and 16.

-	June 14	June 15	June 16
Tare offset	4 712 456	853 686	-1 452 738
Scale factor	14 997,05	-16 127.08	-16 114,87

Table 1: Calibration values for every nights

After that, the two calibration parameters were set and the code was uploaded to the arduino before plugging it into the wall with an arduino alimentation cable. Data acquisition lasted for 15 hours approximately, from 17:00 to 9:00. The sample rate for the night of June was a little bit more than 2 readings/sec whereas it was a little bit more than 1 reading/sec for the nights of June 15 and June 16. Also, for a weight value, 5 readings were averaged to reduce noise. This will probably be done in real experience context too, probably more than 5.

In order to calibrate the loadcell, the member function of the *Loadcell.h* library *calibrate_all()* was used to get the tare_offset and the scale_coeff. These value were then used in this arduino sketch. The code is on the github repo.

Results A low-pass filter was used to reduce noise. These graphs were obtained.

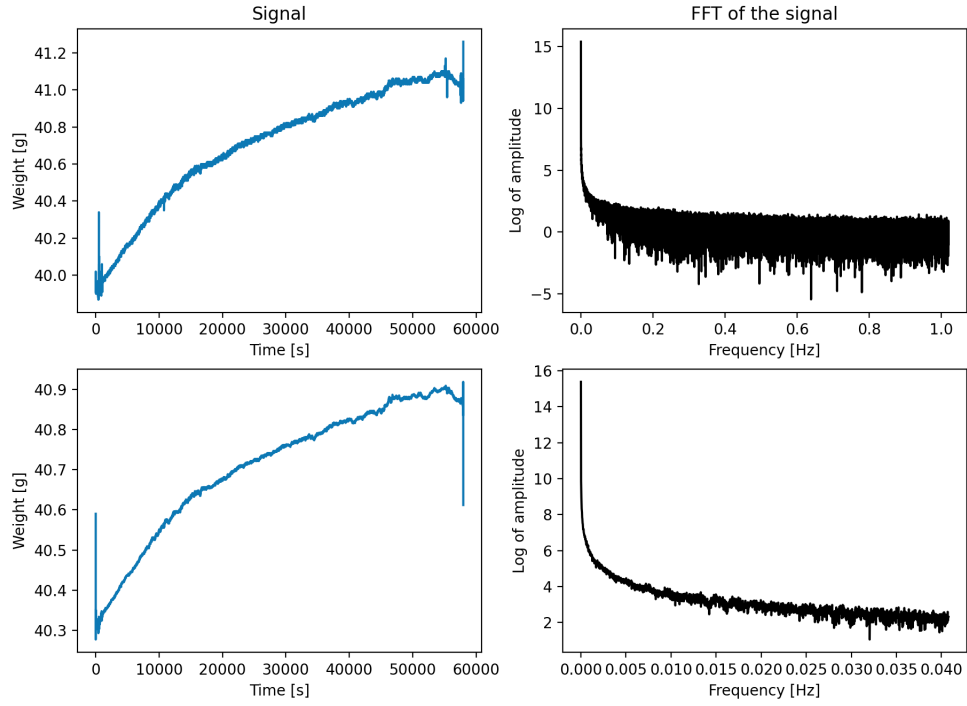


Figure 4: Signal and spectrum for non-filtered (top) and filtered (bottom) for June 14

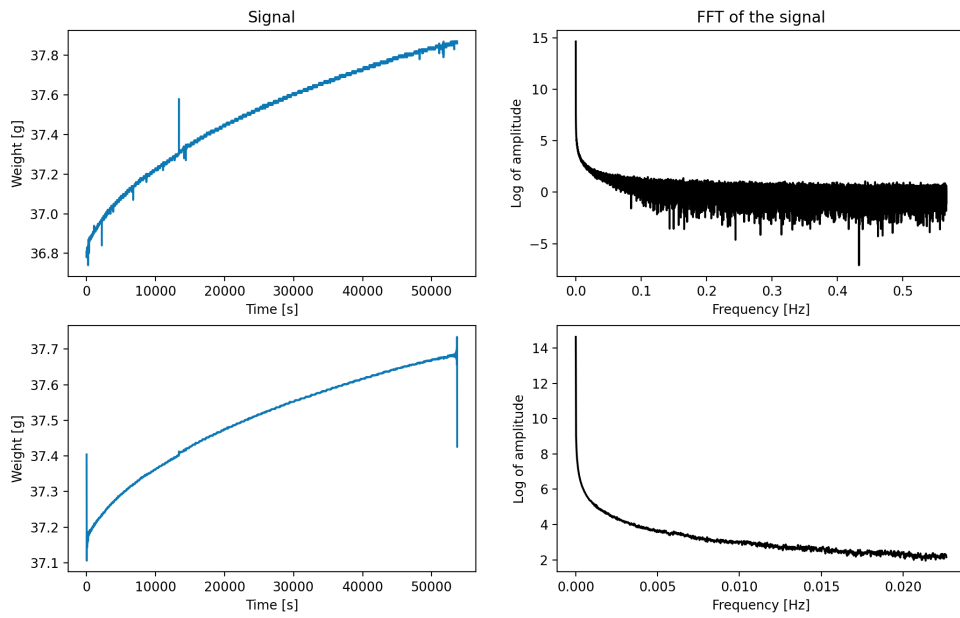


Figure 5: Signal and spectrum for non-filtered (top) and filtered (bottom) for June 15

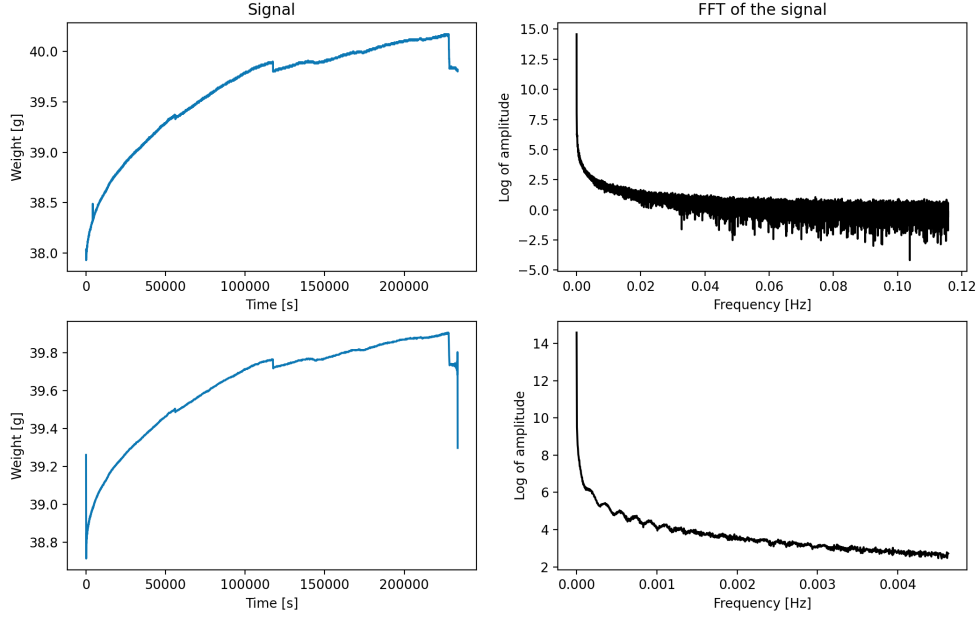


Figure 6: Signal and spectrum for non-filtered (top) and filtered (bottom) for June 16

We can see a clear drift that has a similar behaviour in both tries. But on June 15, a aberrant peak of 500 g was recorded after 15 000 s. This peak has been removed to visualize the drift on the graph. This kind of spontaneous spikes needs to be considered in the code during acquisition or post-acquisition. They are unlikely but not impossible.

Also, we can notice that the initial weight displayed in each graph of each nights does not start at 37 even if the calibration weight was of 37g. Before starting the data acquisition, the loadcell was probably moved after the calibration and this could introduce an error of calibration. But, to visualize the drift behaviour, it wasn't relevant.

2.3 Characterization of the drift with correlation with humidity and temperature with a better 3D printed support and plate at my desk (variable temperature), June 19 2023

Goal Determine the behaviour of the drift of the TAL 221 Loadcell 100g through time and determine if temperature and humidity influence the reading.

Motivation Same as previous.

Methods A similar setup was used with a few upgrades. First of all, better plastic supports were 3D printed (see picture) to have a better stability and something closer to what will be used in the cage.

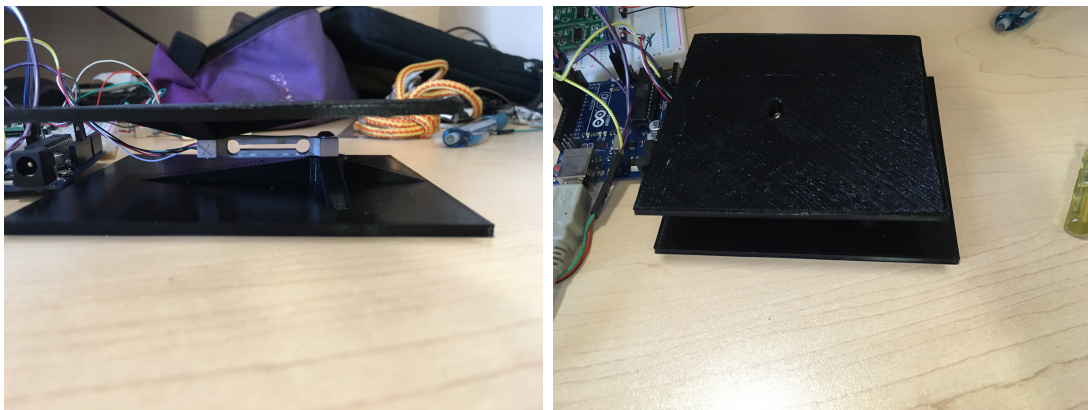


Figure 7: Pictures of support plates (side and front view)

Also, a DHT22 temperature and humidity sensor was used. The circuit is the same as for previous test with the DHT22.

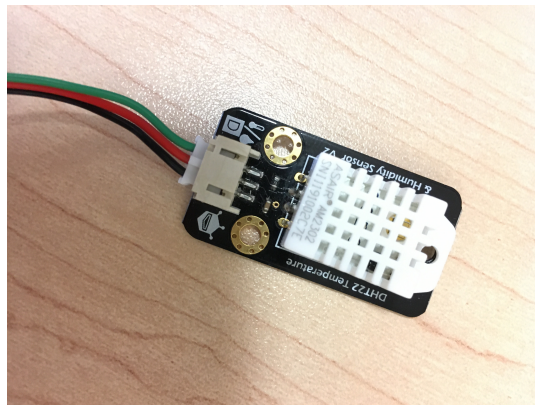


Figure 8: DHT22 sensor

The red wire (VCC) was plugged in the 5V port, the black one in the ground and the green one in a digital pin to read the signal. The arduino code used is on the github repo.

The data acquisition lasted 40 hours and 40 weight-readings were averaged for one measurements. 50 readings for the calibration as displayed in the arduino code. The same starting process was used. The calibration factors are the following:

-	June 19
Tare offset	-1 505 024
Scale factor	-16 070,46

Table 2: Calibration values for every nights

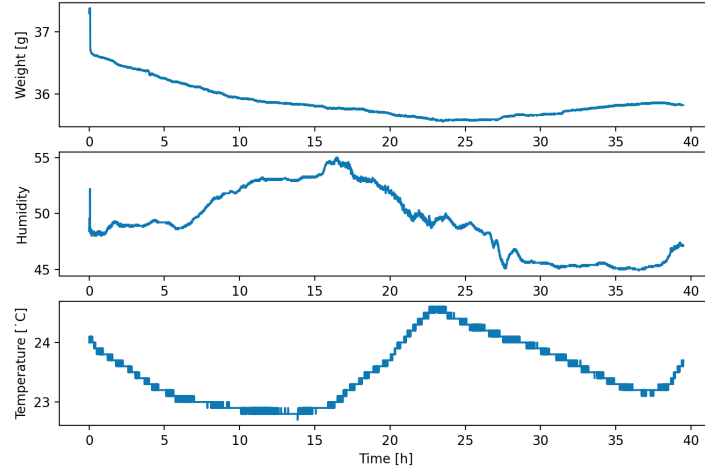


Figure 9: Signal, temperature and humidity for June 19

Results We can see that the drift behaviour is not the same as before. It is the opposite. We can't really determine a correlation with the temperature or humidity. The behaviour of the drift can show that the initial hypothesis of the heating up of the circuit might be incomplete.

2.4 Characterization of the drift with correlation with humidity and temperature with a better 3D printed support and plate in the basement where mices are (supposedly stable temperature), June 21 2023

Goal Determine the behaviour of the drift of the TAL 221 Loadcell 100g through time and determine if temperature and humidity influence the reading.

Motivation Same as previous.

Methods Same as previous but raw readings was also recorded. The same code (on git repo) was used but slightly different calibration parameters were used. The acquisition lasted more than 110 hours.

-	June 19
Tare offset	-1 505 024
Scale factor	-16 070,46

Table 3: Calibration values for every nights

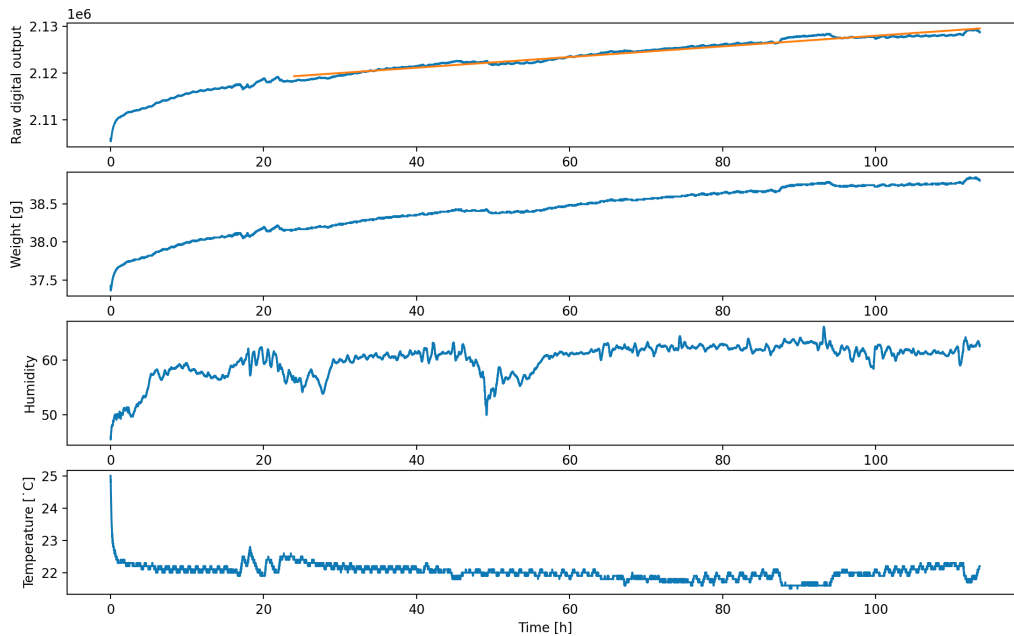


Figure 10: Signal, weight, temperature and humidity for June 19

Results

We can see that we obtained the same behaviour that we had in the first tests. After 24 hours, the drift seems to be constant. A linear curve was fitted (orange) in the raw data to characterize it. The slope is 114.18 and it will be used in the next test to try a simple correction. This slope was determined with `scipy.optimize.curvefit()` in python.

Also, we can look at how the temperature is more constant in the basement than at my desk at the sixth floor. We can also see that a small change in temperature has a direct effect on the raw reading of the loadcell. It shows that temperature has an influence on the Wheatstone bridge equilibrium and consequently on the reading.

2.5 Characterization of the drift with correlation with humidity and temperature in the basement and a first drift correction try, June 27 2023

Goal Determine if a linear drift correction after 24 hours is reproducible for different data acquisition on different night.

Motivation Correct the drift

Methods Same as previous but a drift correction on the raw reading was used. After 24 hours, the following correction was applied where time is in hours.

$$\text{corrected} = 114.18(\text{time} - 24) \pm \text{raw} \Big|_{24h} \quad (1)$$

The $\pm$ depends on the orientation of the wiring of the loadcell and the Wheatstone bridge equilibrium.

-	June 21
Tare offset	1 428 674
Scale factor	-12 529,68

Table 4: Calibration values

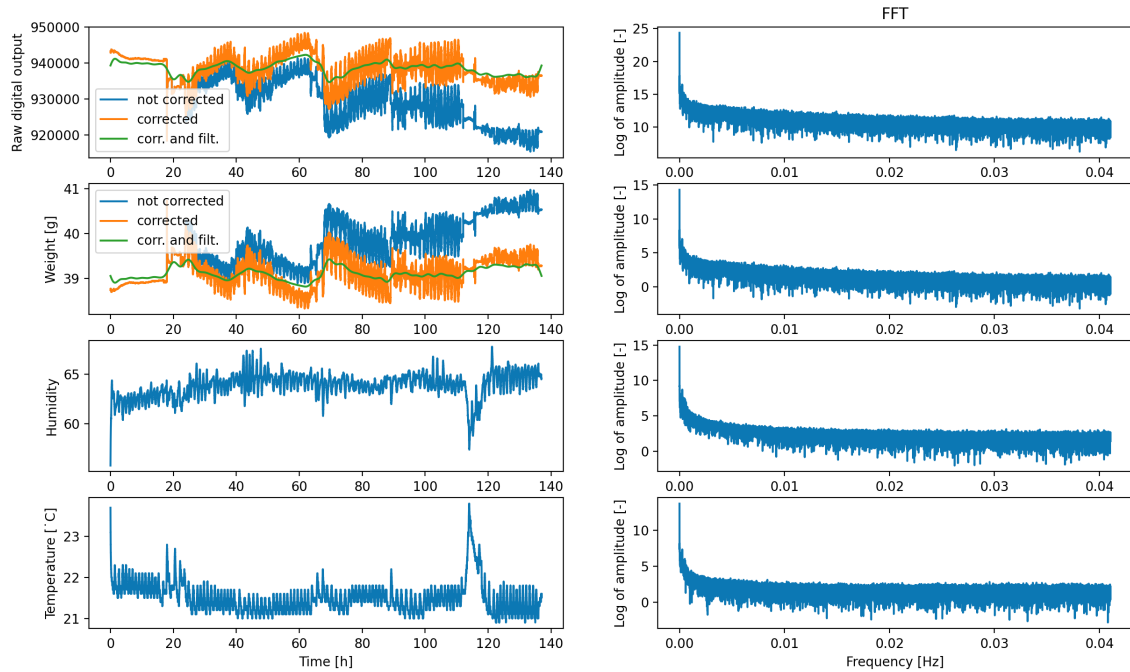


Figure 11: Signal, weight, temperature and humidity and their FFT for June 27th

Results First of all, we can see that we don't have the same behaviour for the raw output and the weight. It is much more noisy than before and the behaviour is not the same as the previous test that was realized in the exact same conditions in the basement.

But, after many tests and screwing and unscrewing the 3D printed support threads began to be loose. I wasn't able to tighten the screws so the support was loose and could cause some mechanic noise. To see if there was any kind of vibration around the loadcell, I used an FFT of all 4 signals and nothing came out of it.

After discussion during the lab meeting, we figured out that the drift might be mechanical and not from electronics, or at least not as important as predicted. Creeping of the loadcell itself and the loadcell's supports might be responsible of the drift. Creeping is the flexion of a material under a pressure and has a similar behaviour to the drift observed. At first, there is a rapid creep in response to the impulsion created by the weight. Throughout time, the flexion keeps going and follows a linear tendency. This is similar to what was observed before in previous experiments.

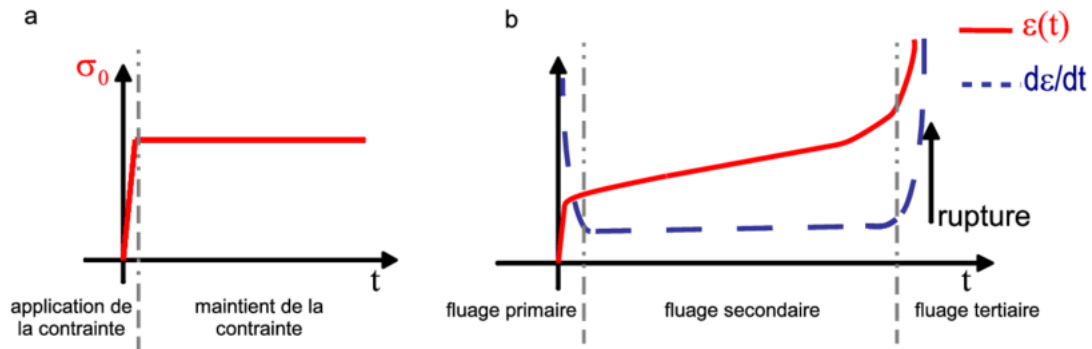


Figure 12: Creep of a material

The linear correction applied on the raw output seems to have attenuated to drift but the noise of the signal is too heavy to conclude anything. But, we can see that the slope of 114.18 is possibly reproducible for different acquisitions.

A low-pass filter was also used to clean-up the corrected signal to observe the behaviour more clearly.

2.6 Characterization of the drift of raw output with correlation with humidity and temperature without support at my desk, July 4

Goal Determine if the loadcell drift is linked to the creep of the 3D printed supports and the loadcell itself or to other parameters.

Motivation Correct the drift

Methods Same as previous but the loadcell was simply on the desk without any 3D printed support to isolate the system from mechanical creep of the plastic. The code is on the github repo.

No calibration was needed to get the raw output and it would have been hard to do so without supports. But, to evaluate the importance of the drift and see if it is negligible when it comes to weight drift, weight was calculated from the raw output after the acquisition (during acquisition would need a calibration). An approximative slope and approximative offset were needed to perform these calculations. A slope of 15 000 was used because it is a fair guess from what was used in previous tests. The offset was calculated with an average of the first 50 readings. The weight was calculated with this equation

$$\text{mass} = \frac{\text{reading} - \text{offset}}{\text{slope}} \quad (2)$$

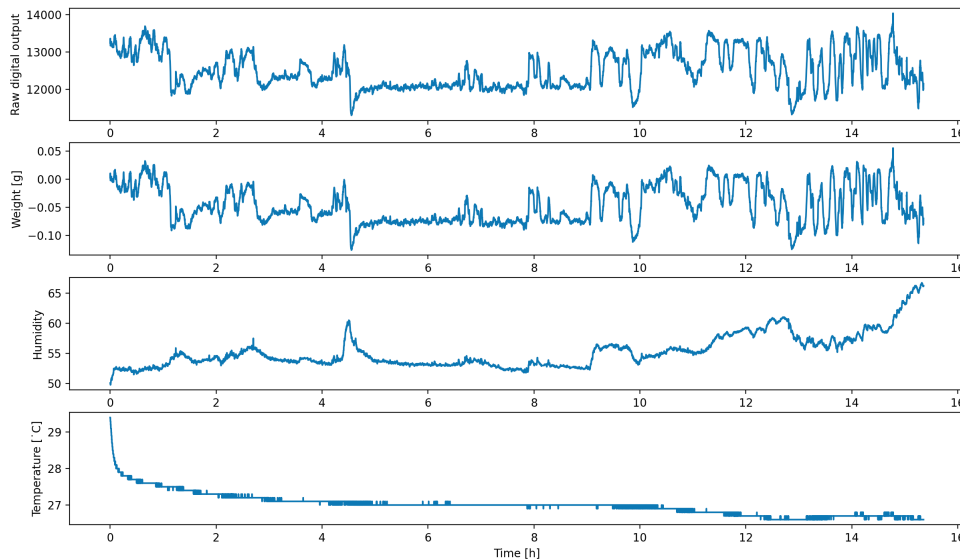


Figure 13: Raw output, approximation of weight, humidity and temperature for July 4th

Results In this test, drift doesn't have the same behaviour as before. It only oscillates. This oscillation are of about 0.15 g peak to peak, which is not as significant as in previous tests. This proposes that the creep hypothesis might be right (or partially)

but more tests need to be executed before concluding anything. Temperature was constant if we ignore the first hour so it is hard to link it to the drift behaviour, the oscillations.

2.7 Characterization of the drift of raw output with correlation with humidity and temperature without support at my desk, July 5,6,7

Goal Determine if the loadcell drift is linked to the creep of the 3D printed supports and the loadcell itself or to other parameters. See if the drift is similar for many loadcells if used simultaneously.

Motivation Correct the drift

Methods The circuit was the same as previous, but two loadcells with an HX711 were added to it. Both of the HX711 were connected to the Arduino Uno on different digital pins to read them. The circuit was the following

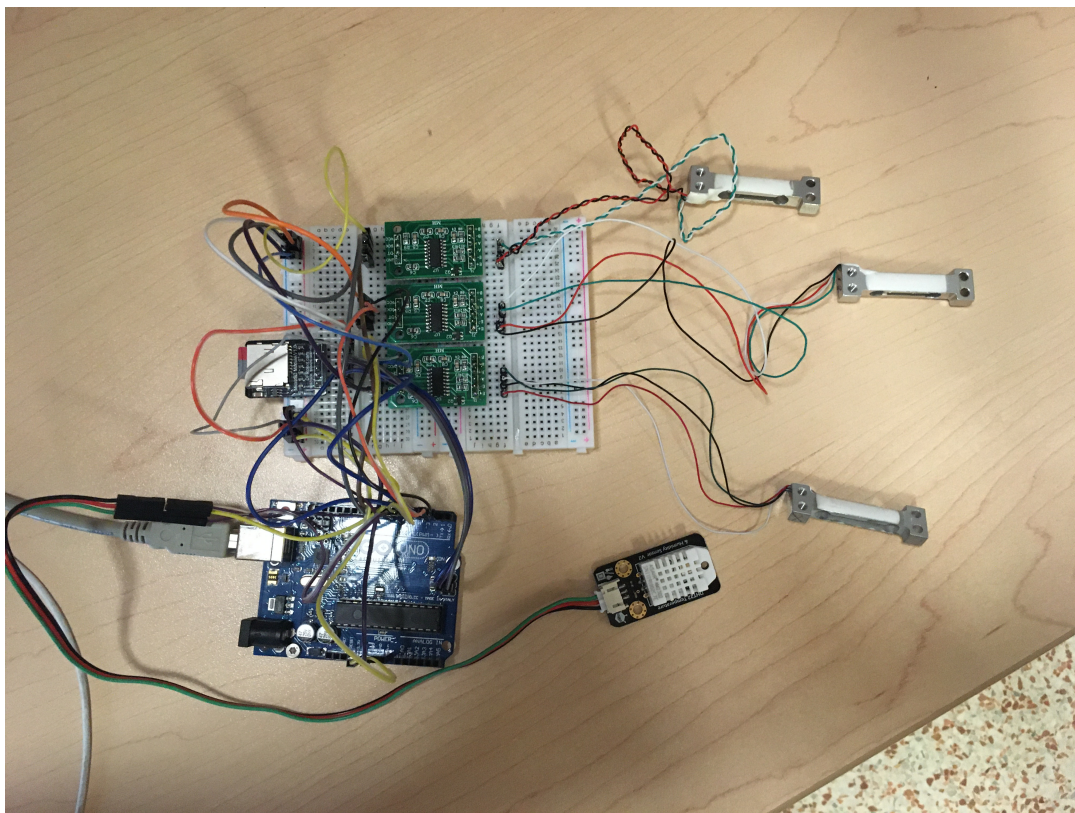


Figure 14: Caption

The code was slightly different to handle many loadcells. It's on the github repo.

As previously, no calibration was needed to get the raw output and it would have been hard to do so without supports. But, to evaluate the importance of the drift and see if it is negligible when it comes to weight drift, weight was calculated from the raw output after the acquisition (during acquisition would need a calibration). An approximative slope and approximative offset were needed to perform these calculations. A slope of 15 000 was used because it is a fair guess from what was used

in previous tests. The offset was calculated with an average of the first 50 readings. The weight was calculated with this equation

$$\text{mass} = \frac{\text{reading} - \text{offset}}{\text{slope}} \quad (3)$$

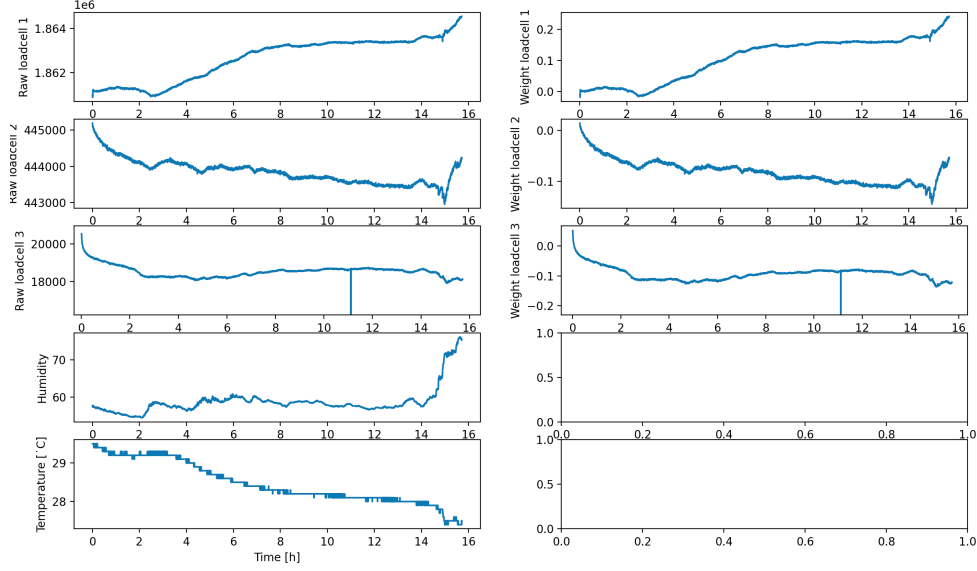


Figure 15: Raw output, approximation of weight, humidity and temperature for 3 load-cells, July 5th

Results For the night on July 5th, we have a drift behaviour that looks like we had with 3D printed supports even if the difference between start and end is smaller without supports. The drift suggests a link with temperature, the constant temperature, then a drop and constant again, etc. But, loadcell 1 drifted towards a higher value in opposition to the two others.

Also, loadcell 3 presents a spike around hour 11, it is not rare to see this kind of event. It shouldn't be hard to correct this.

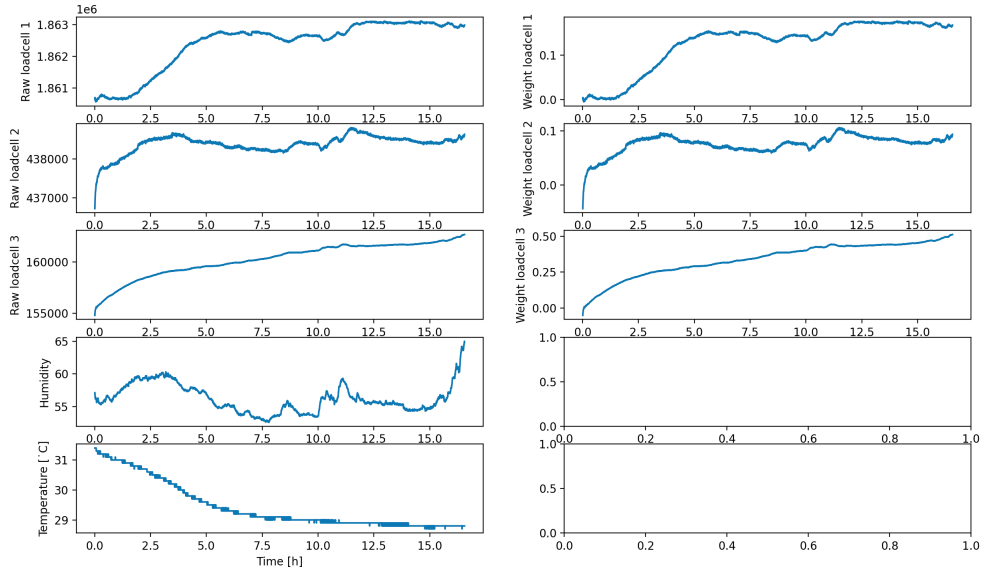


Figure 16: Raw output, approximation of weight, humidity and temperature for 3 loadcells, July 6th

For July 6th night, all loadcells had a similar behaviour that could be explained by temperature and humidity. But, we can't explain why temperature is sometimes increasing the reading or decreasing the reading if we compare all the tests done.

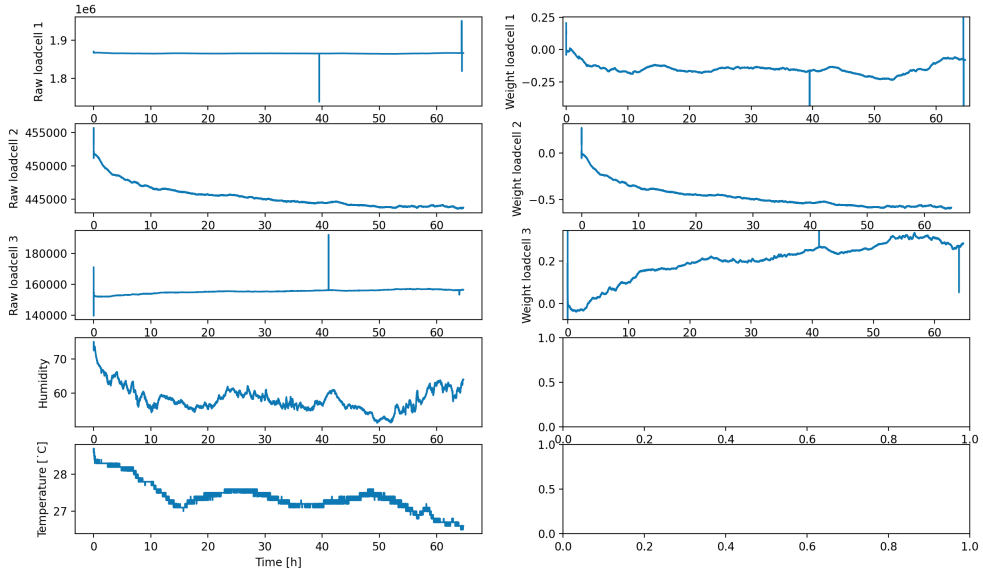


Figure 17: Raw output, approximation of weight, humidity and temperature for 3 loadcells, July 7th

On July 7th, all loadcells had different drift patterns. Loadcell 1 remained fairly constant, loadcell 2 dropped rapidly at the beginning and decreased slowly after and loadcell 3 gradually increased. Consequently, it is hard to find a link between

reading and temperature/humidity. We can also see a few spikes for loadcell 1 and 3.

Notes It is hard to find a drift pattern that is reproducible for all loadcells. Each loadcell is different and the drift behaviour changes for different acquisitions. It makes it really hard to characterize it and find a way to correct it. Still, temperature of the resistances inside the loadcells that form the Weathstone bridge is important. It is written on the datasheet and many websites/people online say it. But, with the DHT22, the temperature sensed is the air temperature, not the circuit temperature. A RTD temperature sensor could be used like in <https://www.phidgets.com/?view=articles&article=LoadCellCorrection#:~:text=For%20the%20simplest%20temperature%20compensation,ll%20call%20it%20Mtemp> to get the temperature of the loadcell.

Self-heating of the circuit is also a possibility. This could be compensated with a RTD temperature sensor.

2.8 Drift conclusions

In the precedent sections, a lot of tests were done to try to characterize it and see if it is significant for TAL221 Load cell 100g. It was shown that the drift is significant and cannot be neglected for the weighing of mice, a high precision is required. It wasn't rare to see a drift of over 1 g in a few days. The behavior was also unpredictable. The drift wasn't always in the same direction either. A drift correction was also tempted with a linear fit after 24 hours but it wasn't conclusive. No characteristic drift behavior was found. Furthermore, two different load cells in the same condition didn't have a similar drift behavior. The characterization of the drift behavior was not impossible.

Many parameters seem to influence the weight measurement/drift.

Temperature It influences the weight measurement since the load cell uses a Wheatstone bridge made of resistors. Resistors, in general, are heat-sensitive and therefore the temperature will affect their resistance value. It was seen in previous tests (figure 10) that a variation in temperature was influencing the weight reading. A rapid temperature variation created a rapid variation of weight. But, the relation between temperature and weight wasn't constant. In some tests, a increase in temperature increased the weight measurement but sometimes it was the opposite. In this experiment <https://www.phidgets.com/?view=articles&article=LoadCellCorrection#:~:text=For%20the%20simplest%20temperature%20compensation,11%20call%20it%20Mtemp>, they present a load cell correction based on the temperature.

Humidity With the DHT22 thermometer, I was able to get a humidity reading. No strong correlation was found with load cell drift. We couldn't come up with a model based on this parameter.

Mechanical creep A load cell bends according to the weight of the object on it. But, with a platform to create a scale, the force created by the object is first transmitted to the platform before being transmitted to the load cell. The way the force is transmitted by the load cell needs to be a constant as possible through time to minimize drift. But when exposed to a weight for a long period of time (many hours), the 3D printed plastic platforms are slowly creeping like explained in a previous experiments. This deformation creates a drift of the measurements since it is changing the rest state of the platform, the offset. It is possible to reduce creep by having small platforms made of a very stiff material. But for our purpose, the platforms can't be very small. To stiffen them, they should be printed with a high infill density.

Solution

Since drift seems unpredictable, it would be hard/impossible to correct it. But, it is possible to take advantage of the tiling of load cells and the slow speed of the drift to minimize it.

When weighing an object, we are only interested on the difference of output before and after we put the object on the scale. It is not important what was the weight value before

and after, only the difference is useful. To use this idea to compensate the load cell, we need to make two assumptions:

- A 20g object will create the same jump in output (the same strain gauges resistance variation). In other words, the relation between weight and jump in output is linear independently of what is the offset.
- The drift of the weight measurement is due to the drift of the load cell only (or in majority) and not to the drift of the scale coefficient. With this assumption, it is possible to compensate the drift by adjusting the offset automatically when the scale is empty.

With a tiling of load cells, all the load cells but one will have a mouse on it. It is possible to detect on which load cell the mouse and then do a measurement of the empty ones to set the value to their new offset. When the mouse will move, we can do the same thing with the other load cell. This way, all the load cell can keep up with the drift.

I have tested this idea a little bit and it seems to be satisfying. The drift is attenuated from 1g for a few days to 0.03g. More tests need to be done to confirm this result. The test was done in perfect condition without a mouse, no cage bedding, etc.

A code example implementing this drift correction strategy is available on the github repo: https://github.com/DCC-Lab/IntelligentCage/tree/master/Loadcell/ArduinoLibraries/Sketch_examples/WeightReadingWithRecalibrationSketch